

# Ideal Reduced Products and a Diagonal Obstruction to BCP

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## Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital  $C^*$ -algebra  $C(K)$  has BCP/UBCP exactly when  $K$  has a countable  $\pi$ -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$  of  $K$  is called a  $\pi$ -basis (weak basis) for  $K$  if every nonempty open subset of  $K$  contains some  $U_\alpha$  in  $\mathcal{U}$  [37, Page 15].

In the setting of commutative  $C^*$ -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

**Corollary 3.9.** *Let  $\mathcal{A}$  be a unital commutative  $C^*$ -algebra. Then the followings are equivalent:*

- (1)  $\mathcal{A}$  has the BCP;
- (2) the pure state space  $\mathfrak{P}(\mathcal{A})$  has a countable  $\pi$ -basis;
- (3)  $\mathcal{A}$  has the UBCP.

In particular, there exist commutative  $C^*$ -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

**Question 3.** *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those  $C^*$ -algebras  $\mathcal{A}$  having BCP/UBCP.*

## Definitions

An ideal  $\mathcal{I}$  on  $\mathbb{N}$  is called proper if  $\mathbb{N} \notin \mathcal{I}$ . A set  $S \subseteq \mathbb{N}$  is  $\mathcal{I}$ -positive if  $S \notin \mathcal{I}$ .

**Definition 1.** *A proper ideal  $\mathcal{I}$  on  $\mathbb{N}$  has the countable disjoint-refinement property if, for every sequence  $(S_k)$  of  $\mathcal{I}$ -positive subsets of  $\mathbb{N}$ , there are pairwise disjoint  $\mathcal{I}$ -positive sets  $T_k \subseteq S_k$ .*

Let  $(E_n)$  be a sequence of normed spaces. Define

$$c_{\mathcal{I}}(E_n) = \{x = (x_n) \in \ell_\infty(E_n) : \{n : \|x_n\| \geq \varepsilon\} \in \mathcal{I} \text{ for every } \varepsilon > 0\}.$$

This is the usual  $c_0$ -sum when  $\mathcal{I}$  is the ideal of finite subsets of  $\mathbb{N}$ . It is closed in  $\ell_\infty(E_n)$ : if  $x^{(m)} \rightarrow x$  in  $\ell_\infty$  and  $x^{(m)} \in c_{\mathcal{I}}(E_n)$ , then for fixed  $\varepsilon > 0$  and large  $m$ ,

$$\{n : \|x_n\| \geq \varepsilon\} \subseteq \{n : \|x_n^{(m)}\| \geq \varepsilon/2\} \in \mathcal{I}.$$

## Main Result

**Theorem 1.** *Let  $\mathcal{I}$  be a proper ideal on  $\mathbb{N}$  containing all finite sets and having the countable disjoint-refinement property. If  $(E_n)$  is any sequence of nonzero normed spaces, then*

$$\ell_\infty(E_n)/c_{\mathcal{I}}(E_n)$$

*fails BCP.*

**Corollary 1.** *Let  $\mathcal{I}$  be as in Theorem 1. If  $(A_n)$  is any sequence of nonzero  $C^*$ -algebras, then the  $C^*$ -reduced product*

$$\prod_{n=1}^{\infty} A_n / \bigoplus_{\mathcal{I}} A_n$$

*fails BCP.*

## Idea of the Proof

The ordinary product quotient  $\ell_\infty(E_n)/c_0(E_n)$  has a Fréchet-limsup norm. The previous diagonal proof attacks different proposed centers along different infinite coordinate sets. For a general ideal  $\mathcal{I}$ , the same proof works exactly when the ideal lets us refine countably many  $\mathcal{I}$ -positive near-norm sets into pairwise disjoint  $\mathcal{I}$ -positive attack sets.

On the attack set assigned to a center, choose unit vectors in the fibers pointing far enough away from that center coordinate. The assembled bounded sequence has reduced-product norm one, while each proposed center sees a positive set of coordinates on which the distance is at least its own norm.

## Proof

We use the standard equivalent form of BCP: if a normed space  $Y$  has BCP, then its unit sphere is covered by countably many open balls

$$B(y_j, \|y_j\|).$$

Thus, to prove failure of BCP, it is enough to show that every countable family of centers misses some point of the unit sphere.

**Lemma 1** (reduced-product norm). *For  $x = (x_n) \in \ell_\infty(E_n)$ ,*

$$\|x + c_{\mathcal{I}}(E_n)\| = \inf_{A \in \mathcal{I}} \sup_{n \notin A} \|x_n\|.$$

*Proof.* Let  $L$  denote the right-hand side. If  $y \in c_{\mathcal{I}}(E_n)$  and  $\varepsilon > 0$ , then  $A_\varepsilon = \{n : \|y_n\| \geq \varepsilon\} \in \mathcal{I}$ . Hence

$$\|x - y\|_\infty \geq \sup_{n \notin A_\varepsilon} \|x_n - y_n\| \geq \sup_{n \notin A_\varepsilon} \|x_n\| - \varepsilon \geq L - \varepsilon.$$

Taking the infimum over  $y$  and then letting  $\varepsilon \downarrow 0$  gives the lower bound.

Conversely, for  $A \in \mathcal{I}$ , define  $y_n = x_n$  on  $A$  and  $y_n = 0$  off  $A$ . Since the support of  $y$  belongs to  $\mathcal{I}$ , we have  $y \in c_{\mathcal{I}}(E_n)$ , and

$$\|x - y\|_\infty = \sup_{n \notin A} \|x_n\|.$$

Taking the infimum over  $A \in \mathcal{I}$  gives the reverse inequality. □

**Lemma 2** (one-coordinate escape). *Let  $E \neq \{0\}$  be a normed space and  $y \in E$ . There is  $u \in S_E$  such that*

$$\|u - y\| \geq \|y\|.$$

*Proof.* If  $y = 0$ , take any  $u \in S_E$ . If  $y \neq 0$ , choose  $f \in S_{E^*}$  with  $f(y) = \|y\|$ . Since  $\|f\| = 1$ , after changing phase if necessary choose  $u \in S_E$  with  $\operatorname{Re} f(u) < -1/2$ . Then

$$\|u - y\| \geq \operatorname{Re} f(y - u) > \|y\| + \frac{1}{2} \geq \|y\|.$$

□

**Lemma 3.** *The ideal of finite subsets of  $\mathbb{N}$  has the countable disjoint-refinement property.*

*Proof.* Let  $(S_k)$  be a sequence of infinite subsets of  $\mathbb{N}$ . Inductively, at stage  $m$ , choose distinct fresh points  $t_{k,m} \in S_k$  for  $1 \leq k \leq m$ , avoiding all points chosen in previous stages and earlier in the same stage. This is possible because only finitely many points are forbidden when each choice is made. Then

$$T_k = \{t_{k,m} : m \geq k\}$$

are pairwise disjoint infinite subsets with  $T_k \subseteq S_k$ . □

*Proof of Theorem 1.* Let

$$\xi_i = a^i + c_{\mathcal{I}}(E_n), \quad i \geq 1,$$

be an arbitrary countable family of centers in the reduced product, where  $a^i = (a_n^i) \in \ell_\infty(E_n)$ . Write  $r_i = \|\xi_i\|$ .

Choose a map  $\pi : \mathbb{N} \rightarrow \mathbb{N}$  such that every fiber  $\pi^{-1}(\{i\})$  is infinite. For each  $k$ , set  $i = \pi(k)$  and define

$$\delta_k = \begin{cases} \min\{r_i/2, 1/k\}, & r_i > 0, \\ 1, & r_i = 0, \end{cases} \quad S_k = \{n : \|a_n^i\| > r_i - \delta_k\}.$$

The set  $S_k$  is  $\mathcal{I}$ -positive. Indeed, if  $S_k \in \mathcal{I}$ , then Lemma 1 would give

$$r_i \leq \sup_{n \notin S_k} \|a_n^i\| \leq r_i - \delta_k$$

when  $r_i > 0$ , a contradiction. If  $r_i = 0$ , then  $S_k = \mathbb{N}$ , which is  $\mathcal{I}$ -positive because  $\mathcal{I}$  is proper.

By the countable disjoint-refinement property, choose pairwise disjoint  $\mathcal{I}$ -positive sets  $T_k \subseteq S_k$ . Define  $b = (b_n)$  as follows. If  $n \in T_k$ , use Lemma 2 in the nonzero space  $E_n$  to choose  $b_n \in S_{E_n}$  satisfying

$$\|b_n - a_n^{\pi(k)}\| \geq \|a_n^{\pi(k)}\|.$$

This is unambiguous because the sets  $T_k$  are pairwise disjoint. For  $n \notin \bigcup_k T_k$ , put  $b_n = 0$ .

Then  $\|b\|_{\ell_\infty} \leq 1$ , and  $\|b + c_{\mathcal{I}}(E_n)\| = 1$ : the norm is at most one, while each  $T_k$  is  $\mathcal{I}$ -positive and  $\|b_n\| = 1$  on  $T_k$ .

Fix  $i$ . If  $r_i = 0$ , then

$$\|b + c_{\mathcal{I}}(E_n) - \xi_i\| \geq 0 = r_i$$

is automatic. Assume  $r_i > 0$ . There are infinitely many  $k$  with  $\pi(k) = i$ . For such  $k$ , on the  $\mathcal{I}$ -positive set  $T_k$  we have

$$\|b_n - a_n^i\| \geq \|a_n^i\| > r_i - \delta_k.$$

Lemma 1 applied to  $b - a^i$  gives

$$\|b + c_{\mathcal{I}}(E_n) - \xi_i\| \geq r_i - \delta_k.$$

Letting  $k \rightarrow \infty$  along the infinite set  $\{k : \pi(k) = i\}$ , we obtain

$$\|b + c_{\mathcal{I}}(E_n) - \xi_i\| \geq r_i = \|\xi_i\|.$$

Thus the unit vector  $b + c_{\mathcal{I}}(E_n)$  lies in none of the open balls  $B(\xi_i, \|\xi_i\|)$ . The countable family of centers was arbitrary, so the reduced product fails BCP.  $\square$

*Proof of Corollary 1.* For  $C^*$ -algebras  $A_n$ , the space

$$\bigoplus_{\mathcal{I}} A_n = c_{\mathcal{I}}(A_n)$$

is a closed two-sided  $C^*$ -ideal of  $\prod_n A_n$ . Apply Theorem 1 to the underlying normed spaces  $E_n = A_n$ .  $\square$

## Scope

For the finite ideal, Lemma 3 recovers the product quotient obstruction  $\ell_{\infty}(E_n)/c_0(E_n)$ . The present formulation isolates the exact combinatorial feature used by the diagonal argument: different centers must be attackable on disjoint positive index sets.

The theorem does not cover ultraproducts over a free ultrafilter. In that case the dual maximal ideal has no two disjoint positive sets. Thus the ultraproduct case, if negative, requires a genuinely different simultaneous escape argument rather than this center-by-center diagonal splitting.

## Verification and Search Bounds

The proof is elementary and uses only the reduced-product norm formula, the Hahn–Banach one-coordinate escape lemma, and the stated disjoint-refinement property. The run indexes were searched for arXiv:2311.14257 together with “reduced product”, “ideal”, “disjoint refinement”, and “ultraproduct”; no existing packet for this ideal-reduced-product BCP obstruction was found.

## Human Review Focus

Review Lemma 1 and the positivity of the sets

$$S_k = \{n : \|a_n^{\pi(k)}\| > r_{\pi(k)} - \delta_k\}.$$

Once those are correct, the rest of the argument is the same diagonal escape mechanism as in the ordinary product quotient.

## References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.